AYA TLIJANI

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PROFILE

Engineering student in Computer Science, specializing in **Data Science and Artificial Intelligence**, with experience in **machine learning research**, **RAG systems and LLM agents**, **data analysis** and **software development**. Seeking opportunities in data science, artificial intelligence and software development.

PROFESSIONAL EXPERIENCE

University of East London – Research Laboratory **Jul 2026 – Nov 2026**
Junior Research Assistant *Remote*

GNN-Based Reinforcement Learning for Project Scheduling | Python, PyTorch, GNNs, Deep Reinforcement Learning

- Developed **graph neural network** policies using **PPO** and self-labeling to solve the Resource-Constrained Project Scheduling Problem (**RCPSP**), benchmarking against classical scheduling heuristics.
- Investigated LFT-anchored learning and conducted reproducible experiments across **PSPLIB** instances with up to 120 activities, statistical testing, and comparative evaluation.
- **Co-authored a research paper** examining fair benchmarking and learned scheduling policies, currently under review.

AMSYS Consulting **Aug 2026 – Dec 2026**
AI Engineering Intern

AI Research & Development, Computer Vision, Deep Learning, Generative AI

- Actively contributing to **AI research and development**, working on advanced **computer vision** and **deep learning** solutions.
- Supporting the development, experimentation, and evaluation of AI models and intelligent systems.

ESPRIM – I5-Assist **Jul 2026 – Aug 2026**
Data Science Intern *Monastir*

Learning Analytics & Competency Assessment | Python, NLP, Machine Learning, scikit-learn, Streamlit

- Developed an **NLP pipeline** using **text classification** and **weak supervision** to assess student competencies from AI interaction logs.
- Built an **interactive dashboard** for competency assessment, evidence-based scoring, student profiling, and learning progression analysis.

GCT Gafsa **Jul 2025 – Aug 2025**
Introductory Internship *Gafsa*

IT support, exploration of the company's network infrastructure, and participation in the development of an e-commerce platform with PrestaShop.

PROJECTS

SciLit – *RAG System for Verified Literature Reviews* | Python, PyTorch, Hugging Face, RAG, LLMs, FastAPI GitHub

- Built an end-to-end **RAG system** over 1,109 NLP papers, combining **hybrid retrieval** (BM25 + dense embeddings, cross-encoder reranking) with a local **LLM** that writes cited literature reviews.
- Added automatic **fact-checking** of every generated sentence against the source papers (NLI): **95%** of sentences pass; fine-tuned **SciBERT** for sentence classification (0.749 macro-F1).

QueryPilot – *Self-Correcting Text-to-SQL Agent* | Python, LangGraph, LLMs, SQL, MCP GitHub

- Built a **LangGraph**-based **LLM agent** that generates SQL from natural-language questions, executes queries, and automatically diagnoses and repairs failures.
- Added a context pipeline (column descriptions, value matching, schema linking) that reached **70% execution accuracy** on the **BIRD Mini-Dev** benchmark with Gemini, **+8 points** over direct SQL generation.
- Implemented a **read-only SQL sandbox**, tested against 16 write and injection attacks, and exposed database tools to other AI assistants through an **MCP** server.

SurgiSight – *Surgical Video Understanding* | Python, PyTorch, timm, OpenCV, FFmpeg GitHub

- Built a **computer vision** pipeline that streams the 75 GB Cholec80 surgical video dataset one video at a time, with GPU decoding and frame cleaning, producing 184k labelled frames (4.72 GB).
- Fine-tuned **ConvNeXt-Tiny** for joint surgical phase recognition and instrument detection on the 32 official test videos: **95.5% tool mAP** and **83.9% phase accuracy**.

EDUCATION

ESPRIT Monastir (ESPRIM) **Sep 2024 – Present**
Engineering Degree in Computer Science, specializing in Data Science and Artificial Intelligence *Monastir*

Preparatory Institute for Engineering Studies of Monastir (IPEIM) **Sep 2021 – Jun 2024**
Preparatory Institute for Engineering Studies *Monastir*

SKILLS, CERTIFICATIONS & LANGUAGES

- **Languages:** Python, SQL, Java, JavaScript, PHP, C, C#
- **Machine Learning & AI:** Deep Learning, Deep Reinforcement Learning, Graph Neural Networks, NLP, Computer Vision, Generative AI, PyTorch, PyTorch Geometric, TensorFlow, scikit-learn
- **LLMs, RAG & Agents:** LangGraph, MCP, Ollama, Hugging Face Transformers, FAISS, BM25, Text-to-SQL, LLM Evaluation
- **Optimization:** OR-Tools, Vehicle Routing (VRPTW), Project Scheduling (RCPSP)
- **Web & Backend:** FastAPI, REST APIs, React, Angular, Symfony, JavaFX, HTML/CSS
- **Data & Tools:** Pandas, NumPy, SQLite, Streamlit, Power BI, Databricks, Jupyter, Git, Docker, pytest
- **Certification:** NVIDIA Deep Learning Institute – Fundamentals of Deep Learning
- **Languages:** Arabic (Native), English (C1), French (B2), Spanish (Basic)